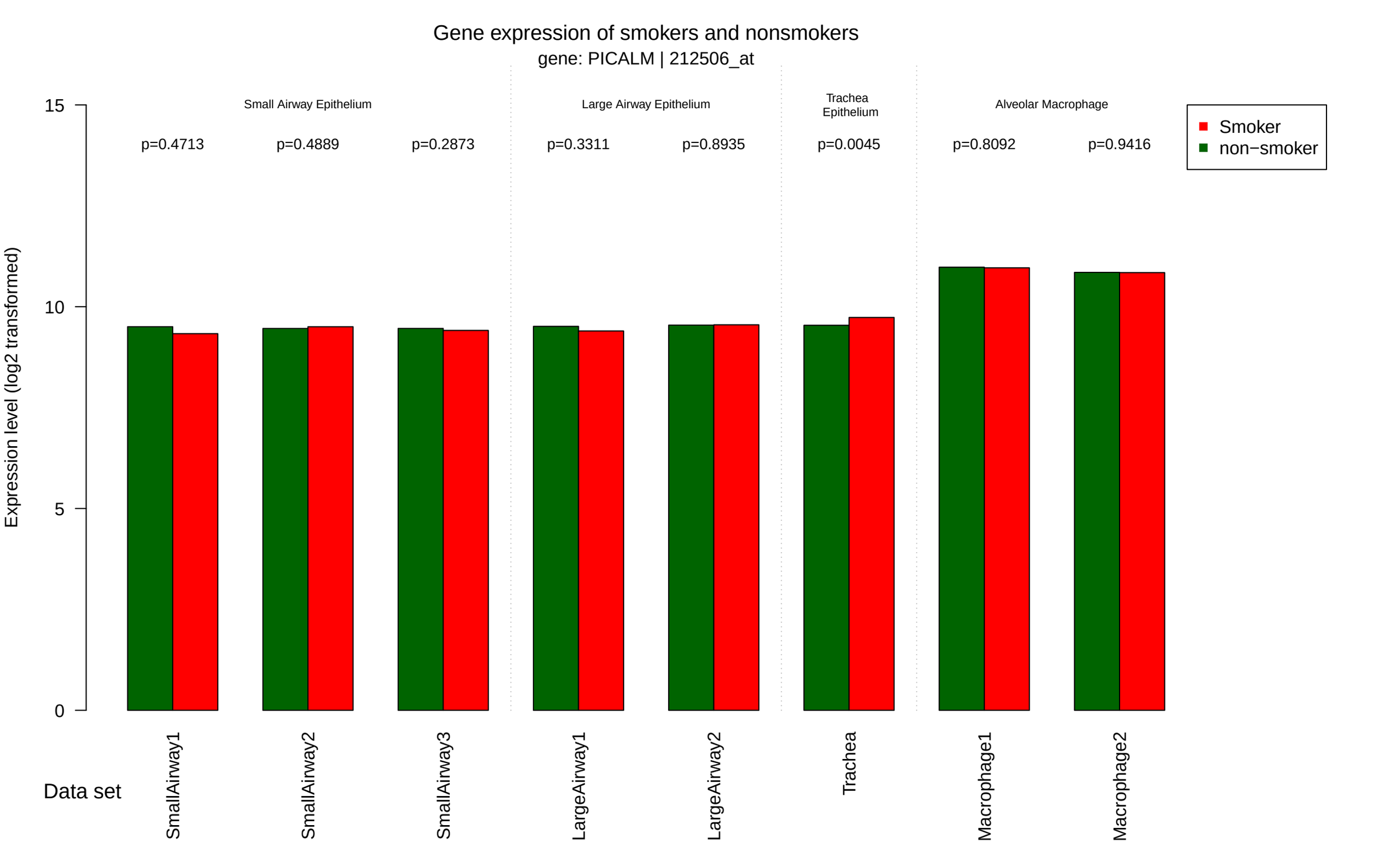
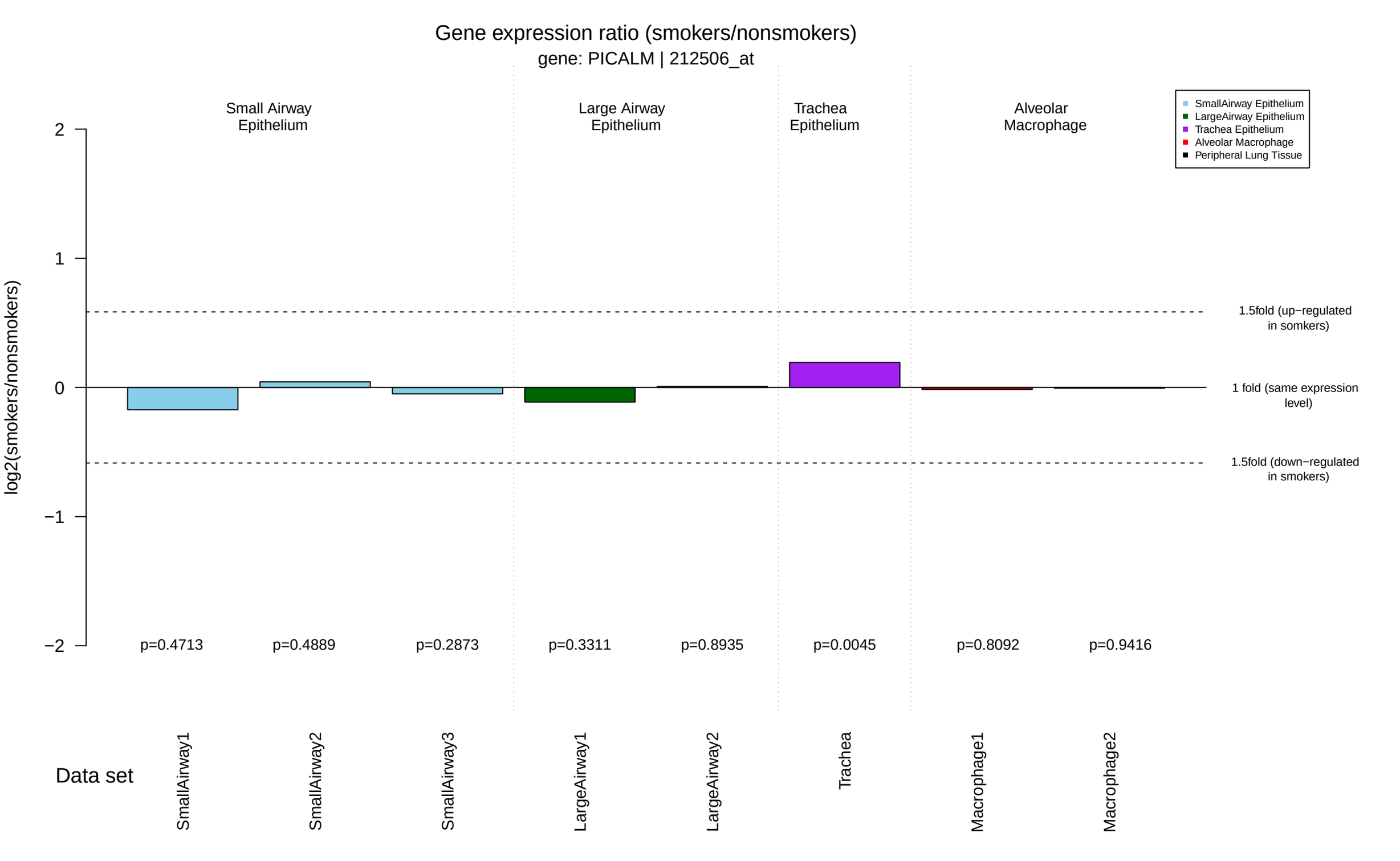
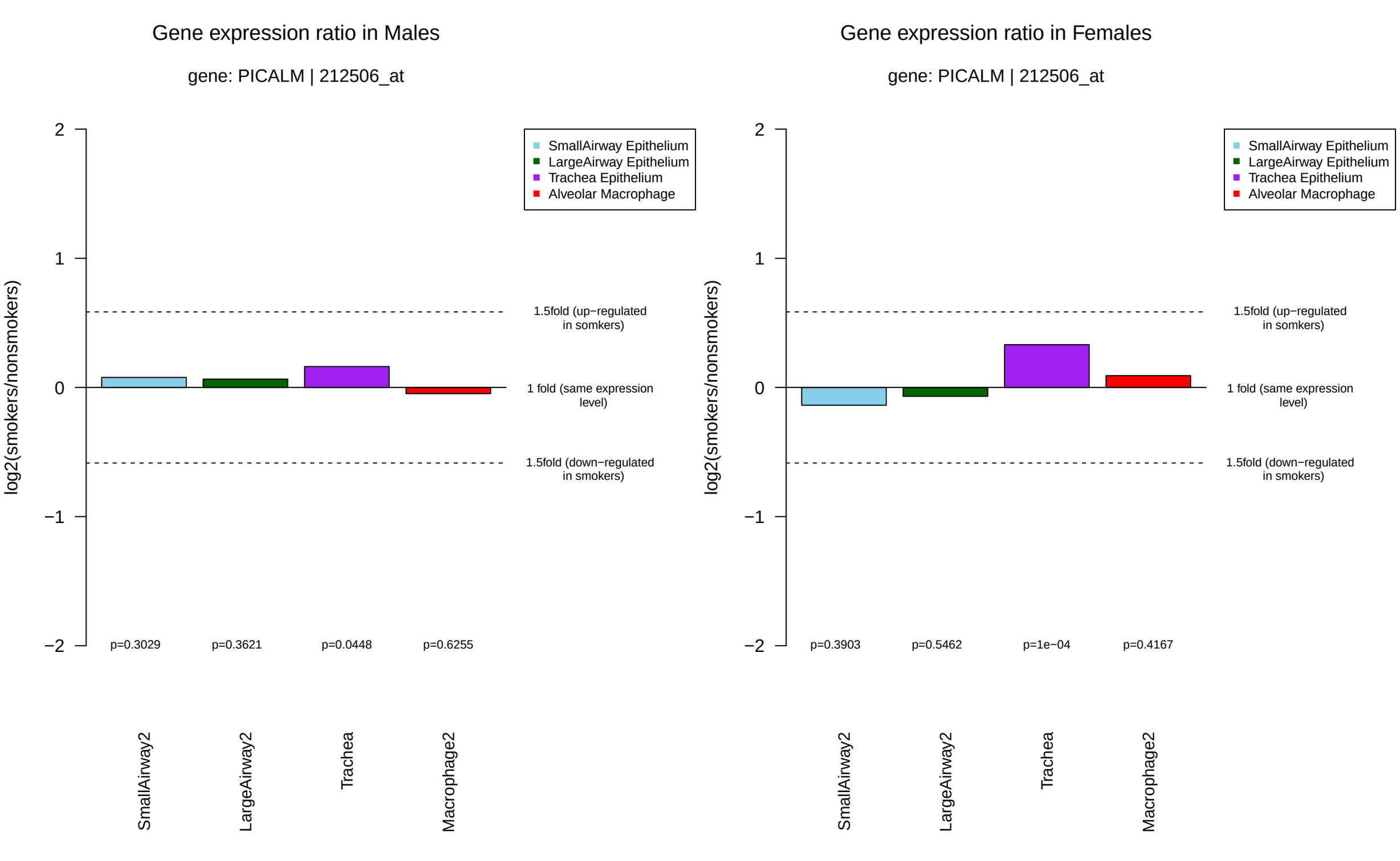
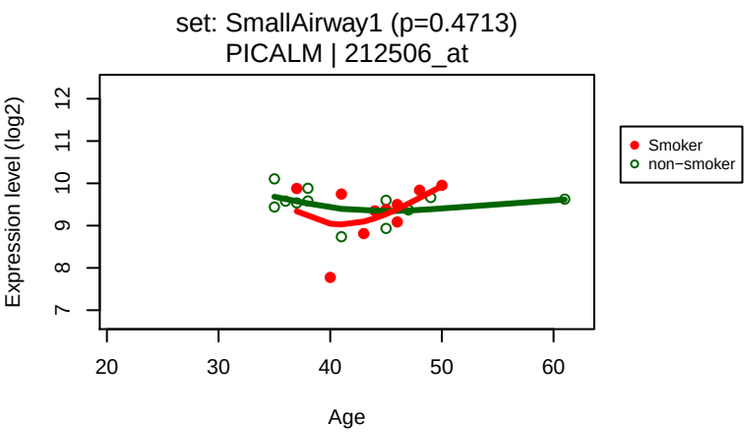
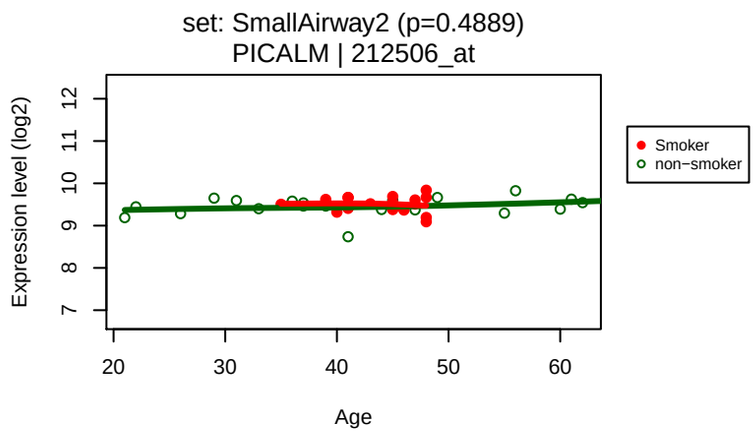
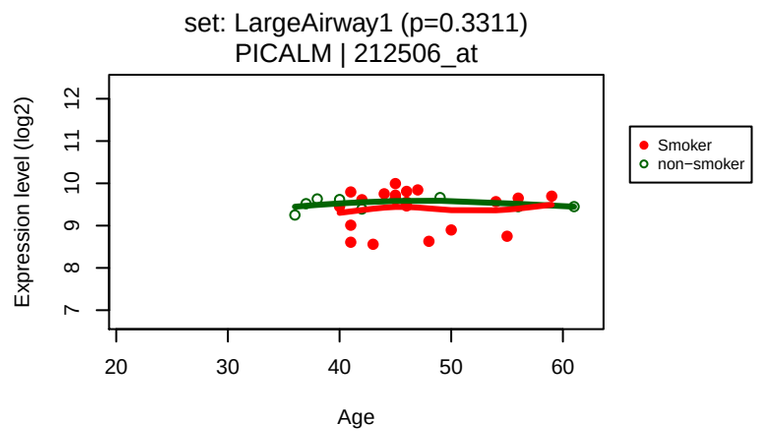

Gene expression of smokers and nonsmokers
gene: PICALM | 212506_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4713
p=0.4889
p=0.2873
p=0.3311
p=0.8935
p=0.0045
p=0.8092
p=0.9416
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

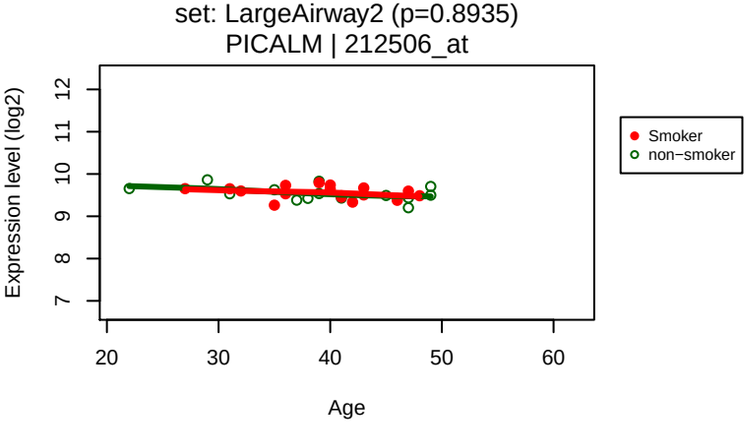

Gene expression ratio (smokers/nonsmokers)
gene: PICALM | 212506_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4713
p=0.4889
p=0.2873
p=0.3311
p=0.8935
p=0.0045
p=0.8092
p=0.9416
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

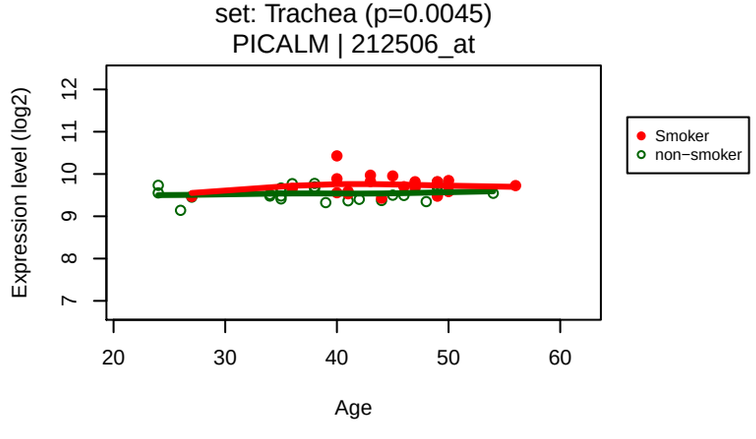

Gene expression ratio in Males
gene: PICALM | 212506_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3029
p=0.3621
p=0.0448
p=0.6255
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PICALM | 212506_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3903
p=0.5462
p=1e−04
p=0.4167
SmallAirway2
LargeAirway2
Trachea
Macrophage2

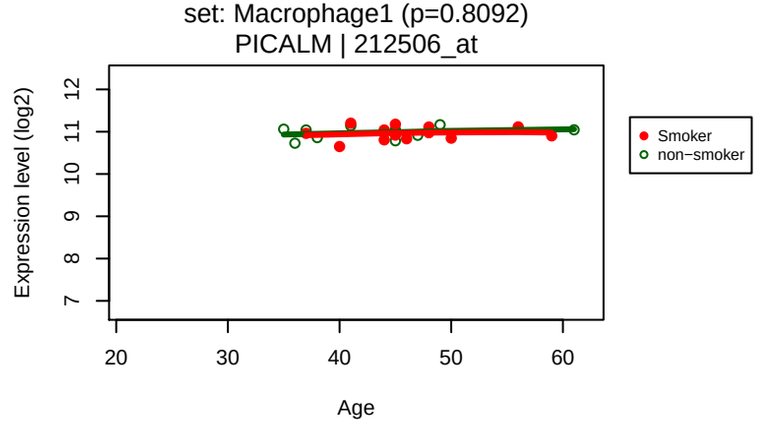

set: SmallAirway1 (p=0.4713)
PICALM | 212506_at
Expression level (log2)
Age
Smoker
non−smoker

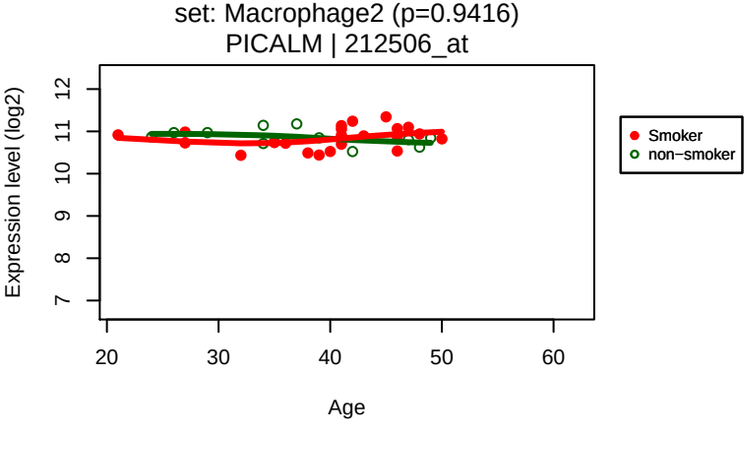

set: SmallAirway2 (p=0.4889)
PICALM | 212506_at
Expression level (log2)
Age
Smoker
non−smoker

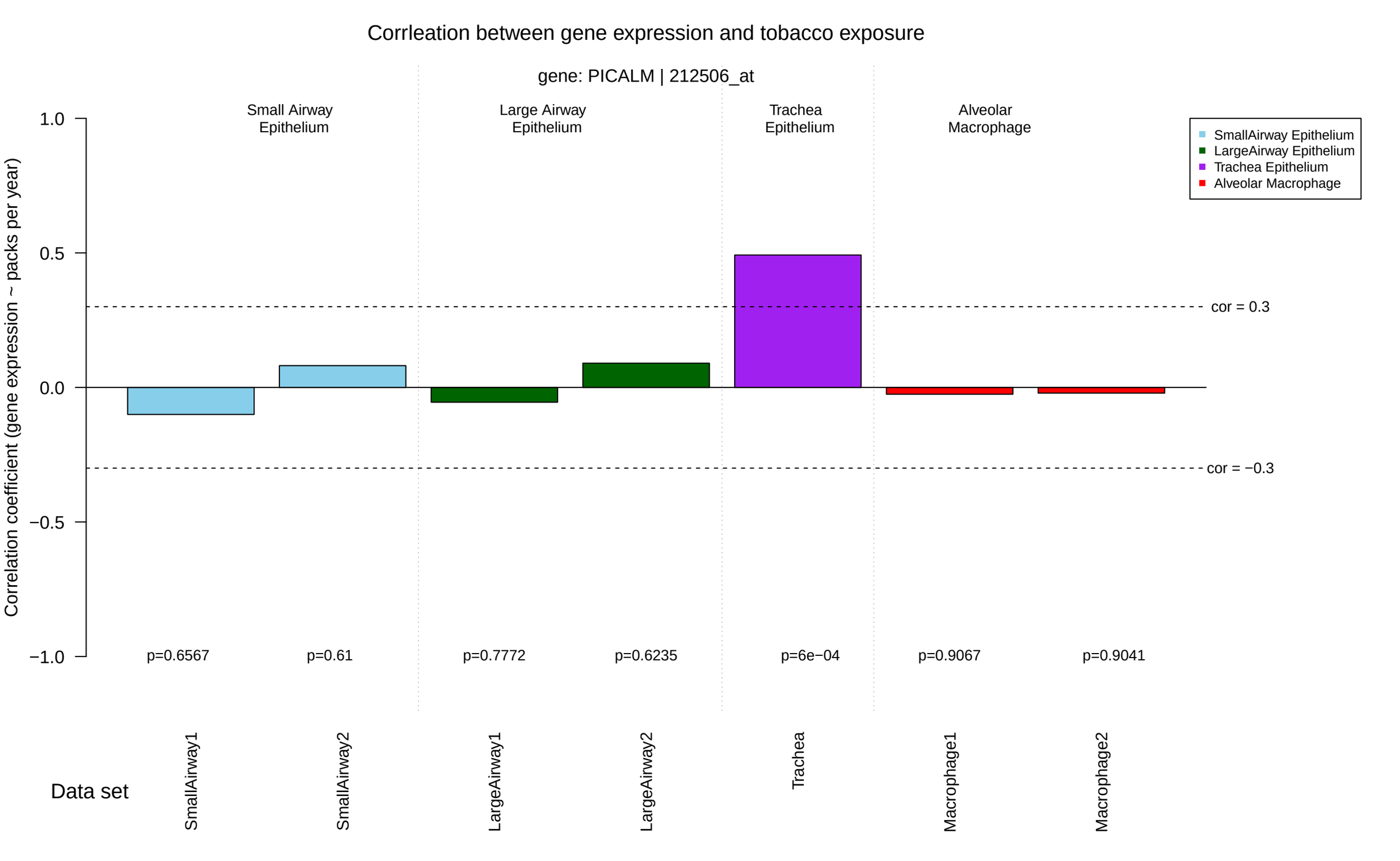

set: LargeAirway1 (p=0.3311)
PICALM | 212506_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8935)
PICALM | 212506_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0045)
PICALM | 212506_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8092)
PICALM | 212506_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9416)
PICALM | 212506_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PICALM | 212506_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6567
p=0.61
p=0.7772
p=0.6235
p=6e−04
p=0.9067
p=0.9041
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2